Atılım Connect

Assoc. Prof. Dr. Cansu Çiğdem EKİN

CMPE/ISE/SE 494 Senior Project

2024-2025 Spring

Team

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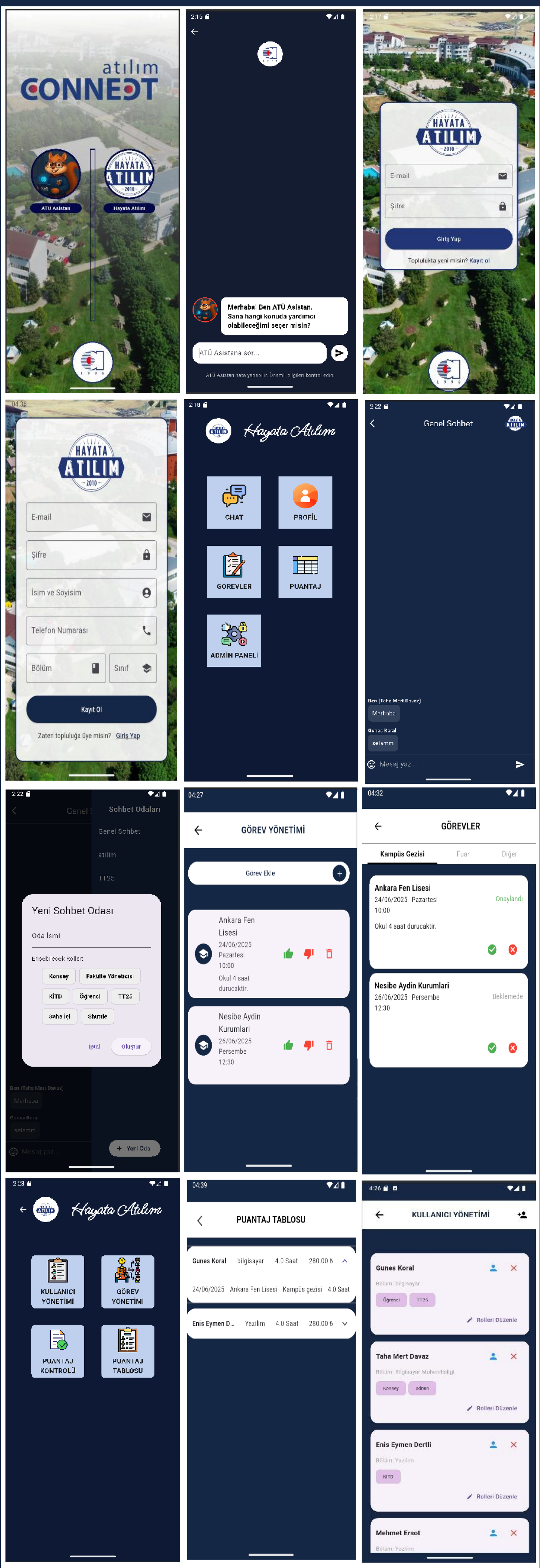
Interfaces



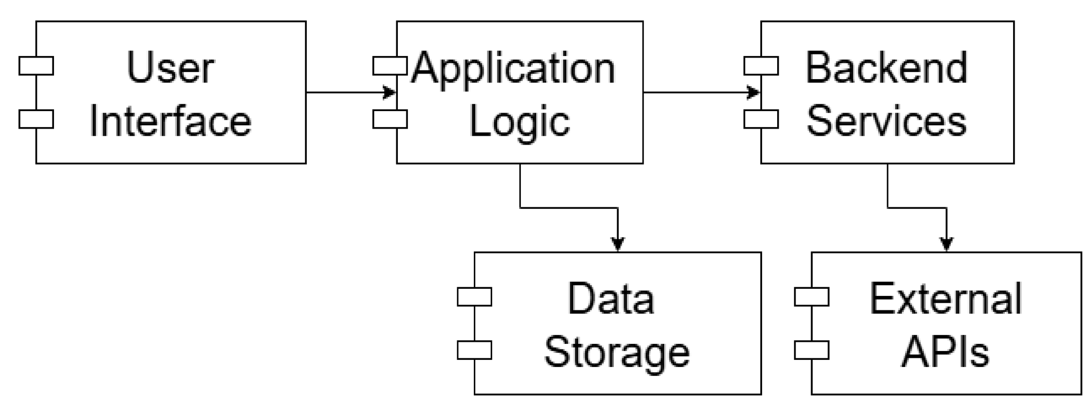
Objective

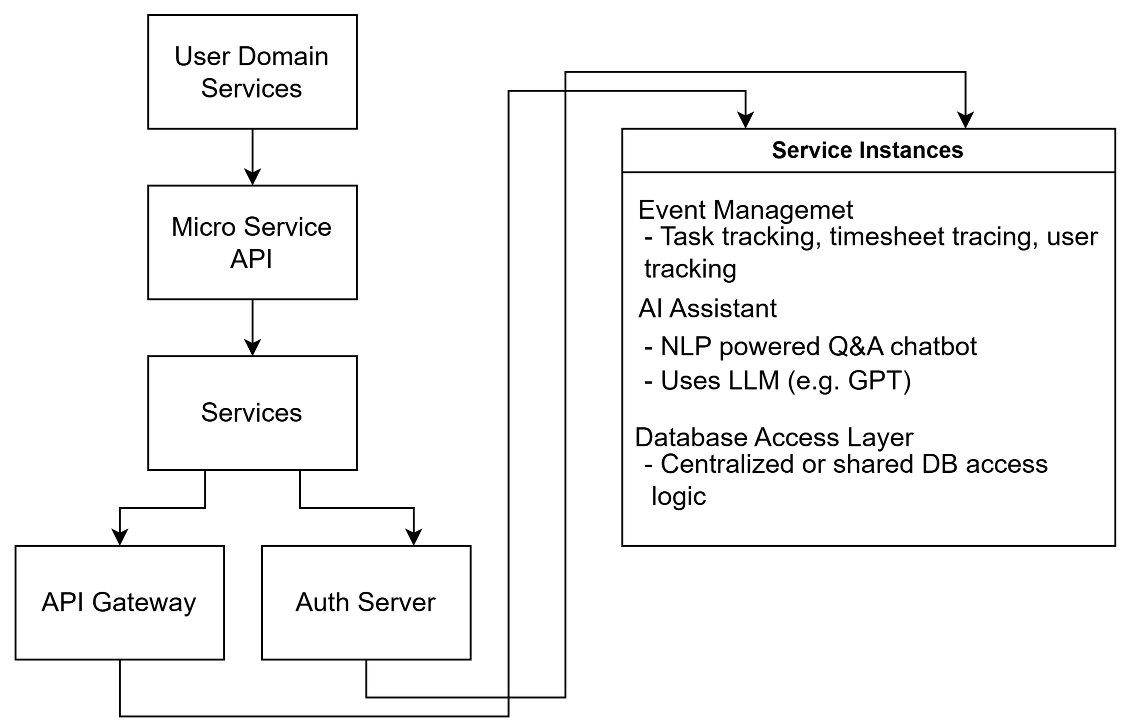
The primary aim of the Atılım Connect project is to improve communication between the Institutional Promotion Directorate and students while easing the adaptation process for new students at Atılım University. The application includes two main components:

Student Coordination Platform for University Promotional Events enables the university promotional team to efficiently assign student roles, track work hours, and maintain smooth communication throughout the events.

AI-based Chatbot to instantly answer students' questions about university regulations, locations, and academic processes.

Design and Architecture





Conclusion

Atılım Connect successfully achieved its objectives by streamlining student-administration communication and improving promotional event coordination. The chatbot effectively delivers real-time responses, and the task tracking system ensures transparent and fair management of student efforts. Despite challenges such as high page complexity and AI integration issues, the project laid a robust foundation for future enhancements. Planned improvements include UI simplification and expanding the chatbot's capabilities based on real user feedback.

Application Areas

Atılım Connect is a mobile application designed to enhance campus life by addressing key needs of the university community. It simplifies university promotion by organizing student participation in events and improving coordination with the Corporate Communications Directorate. A built-in chatbot supports new students by providing instant answers to common questions and clear, easy-to-understand guidance to help them settle in quickly. The app also allows task assignments, tracks time spent, and ensures fair scoring and rewards for student contributions. With real-time messaging and notifications, it enables fast and efficient communication between students and administrators. By incorporating gamification and community tools, the platform boosts motivation, engagement, and a sense of belonging, helping build a more connected university environment.

Methods/Tools

The project was developed following the Agile methodology. The frontend was implemented using Flutter 3.10+ and Dart 3.0+, enabling seamless cross-platform compatibility. For backend services, Firebase Firestore was used for real-time data storage, supported by Firebase Authentication, Cloud Messaging, and Cloud Functions. The AI-powered chatbot was built using a Retrieval-Augmented Generation (RAG) architecture, leveraging OpenAI's GPT models and the LangChain library.

Flutter Firebase Dart OpenAI python Android Studio

References

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